

easier to see in version control systems, such as Git, that track changes programmers make.

Next, we define the constants this program will need:

```
TOTAL_DISKS = 5 # More disks means a more difficult puzzle.

# Start with all disks on tower A:
SOLVED_TOWER = list(range(TOTAL_DISKS, 0, -1))
```

We define these near the top of the file to group them together and make them global variables. We've written their names in capitalized `snake_case` to mark them as constants.

The `TOTAL_DISKS` constant indicates how many disks the puzzle has. The `SOLVED_TOWER` variable is an example of a list that contains a solved tower: it contains every disk with the largest at the bottom and the smallest at the top. We generate this value from the `TOTAL_DISKS` value, and for five disks it's `[5, 4, 3, 2, 1]`.

Notice that there are no type hints in this file. The reason is that we can infer the types of all variables, parameters, and return values from the code. For example, we've assigned the `TOTAL_DISKS` constant the integer value 5. From this, type checkers, such as Mypy, would infer that `TOTAL_DISKS` should contain integers only.

We define a `main()` function, which the program calls near the bottom of the file:

```
def main():
    """Runs a single game of The Tower of Hanoi."""
    print(
        """THE TOWER OF HANOI, by Al Sweigart al@inventwithpython.com

Move the tower of disks, one disk at a time, to another tower. Larger
disks cannot rest on top of a smaller disk.

More info at https://en.wikipedia.org/wiki/Tower_of_Hanoi
"""
    )
```
